The Macroeconomic Limits of AI Compute

A community estimate of whether the build-out can earn its carry — and what it takes

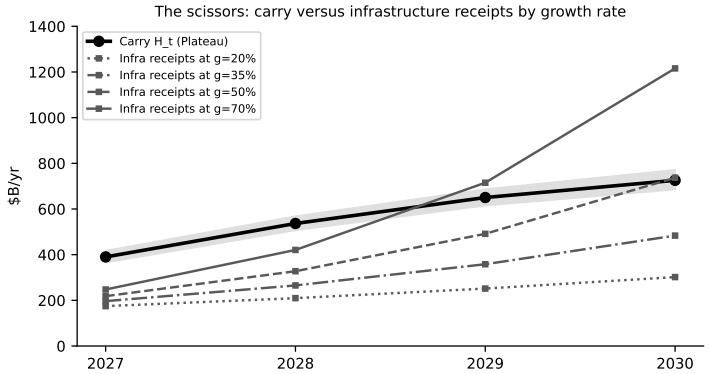
Desmond Fitzpatrick, Ph.D. IBM Systems (retired) September 2026 Summary of v23

Why this matters. Five companies will spend about $780 billion this year on AI data centers — roughly 2.5% of US GDP, the scale of the railway and telecom booms — on the expectation that revenue will arrive to pay for it. Whether it does is not a technology question; it is the largest capital-allocation question in the world economy, and it is being answered mostly by assertion. This project tries to answer it by arithmetic that anyone can check and improve.

What we did. We built a *vintage* model of the installed base: each year's capital spending as a cohort with its own depreciation life (silicon ∼4 years, buildings ∼20), energization lag, grid and siting constraints, and a return on net book. That gives the annual *carry* the base must earn. We then estimated demand two independent ways — top-down, from observed 2026 AI revenue grown at a rate g; and bottom-up, from fifteen sector chains, each bounded by seats, budgets, users or ad yields — and asked what fraction of a customer's dollar actually reaches the infrastructure owner after the labs' margins and the leakage to devices, premises and integrators. Every input is a range with a source; the arithmetic propagates the ranges honestly (independent uncertainties add in quadrature, weighted by share); and the answer is a distribution, not a number.

What we found.

- The carry ramps from ∼$390B in 2027 to ∼$725B in 2030 if spending merely plateaus. No plausible demand growth covers 2027–28: those years are a sunk deficit under every path.
- Breaking even in 2030 on a plateau needs AI revenue to grow ∼49% a year, sustained, from $179B to ∼$880B. The sector chains deliver ∼33%: growth decays as the big sectors approach their ceilings. The 2030 gap is ∼$260B, 3.5 standard deviations of the propagated uncertainty.
- The bull case is not impossible. It is financeable if AI takes ∼25% of an enterprise software-and-services budget that is no longer growing — money that comes out of seat-priced software and consulting hours. That is why AI revenue can approach a trillion while the software industry shrinks: it is the software industry's own money changing hands, and only a fraction reaches the meter.
- Extending to 2035 does not rescue a plateau; a 35% capex cut, or an operator who cuts in step with realized receipts, turns profitable in 2030–31 and recovers by 2034–35. The outcome is more likely to be decided by a capex decision than by a demand surprise.



What would prove us wrong. Calendar-2028 AI revenue above $400B on the paper's basis, or, by 2030, impairments under $100B with no 30% capex retreat. Interim marker: 2027 revenue near $330B is our path; above $350B favors the build-out.

How to help. The estimate lives in a public register of ∼45 inputs, each with its evidence, range, likely bias and what would improve it. Change one, and a script recomputes everything and reports what moved. Modeled on collaborative mathematics: one object of record, small sourced contributions, machinery that checks them, collective credit. The most valuable contribution is a primary source for a number we could only infer — and the companies concerned could remove most of the inference with a few disclosures they already possess.